الشركة السعودية للكهرباء
Saudi Electricity Company

TES-P-119.35

Revision No. 0

STATION AUXILIARY TRANSFORMER SIZING

REVISIONS

No.	Date	DESCRIPTION	ORIG'D	CERT.	APP'D
1					
2					
3					

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE

This standard covers the guidelines for Sizing Station Auxiliary Transformer used in Transmission Substations of Saudi Electricity Company (SEC) Saudi Arabia.

2.0 AUXILIARY SYSTEM

Station Auxiliary system shall comply with the following requirements:

2.1. For standard AC auxiliary system voltage, refer to TES-P-119.02

2.2. The station auxiliary transformer shall meet the requirments of 51-TMSS-01.

2.3. Vector group shall be Dyn11 or Dyn5 as defined in the SOW / PTS.

2.4 The transformer shall be designed and constructed to withstand at any tap without damage, the thermal and mechanical effects of short-circuit in accordance with IEC 60076 or IEEE Std C57.12.00.

3.0 TRANSFORMER SIZING – TYPICAL CALCULATION

3.1 Sizing Criteria

3.1.1 The ultimate substation configuration shall be the basis for estimating the various loads.

3.1.2 The simultaneous peak loading of the auxiliary transformer excluding power transformer oil filtration socket load shall not exceed 80% its rating; and with oil filtration socket, the load shall not exceed the full rating of the station transformer.

3.1.3 The diversity factor of all lighting load shall be considered as 1.0.

3.1.4 The following loads need not be considered for estimating simultaneous peak loading:

a. Transformer OLTC motors
b. Sump Motor
c. All other short duration motor including CB motor (if hydraulic/pneumatic) etc.

3.1.5 Refer to Table 35-1 and Sub-Tables for typical example of loading calculations

3.1.6 Calculation of AC loads shall be based on Substation actual AC auxiliary loads.

4.0 REMARKS FOR DIVERSITY FACTOR

a) Two feeders are feeding the same panel.

b) Assuming 50% of Indoor or outdoor recepetacles are loaded.

c) Momentary load-sump pump, OLTC motors & transformer Oil filteration socket are not considered per clause 4.1.4.

d) Assuming one receptacle in each group of panels/Lighting pole is loaded continuously.

e) Thermostat controlled space heater load in control, Relay & 13.8kV switchgear panels are not considered since 100% HVAC system load is included, which maintains room temperature of 25˚C.

f) All Cubicle lighting loads are included in the calculation.

g) Duct heater loads are excluded since full load of ACCU is considered.

h) Actual Dry type Transformer load is considered.

Note:
The above remarks are for the diversity factor used in the typical claculation. However the actual calculation shal be carried out per the diveristy factor specified in Scope of Work and Technical Specification (SOW/TS).

Table 35-1: Main ACDB loads

S.No.	Description of Load	Load per Equipment in VA (a)	Quantity (b)	Total Connected Load c = (a)X(b)	Diversity factor (d)	Operating Load (c) X (d)	Remarks Refer Clause 5
	380/220 V, 3Ph, 4-wire Main ACDB Bus Section-1						
1	125V Battery Charger-1	37483	1	37483	0.5	18741.5	a
2	380V Oil Filtration Receptacle -T-601 & T602	131640	2	263280	0	0	c
3	PACU-1 & 2 for 13.8kV switchgear room	97647	2	195294	0.5	97647	a
4	Control panel for HVAC & Duct mounted Electric Heater	21176	1	21176	0.05	1058.8	g
5	Power Panel -1 (PP-1) **(refer to table # 35-2)**	28343.3	1	28343.28	1	28343.28	
6	63 Amp Power Socket near TRF. T-601,602	41465	2	82930	0	0	c
7	63 Amp Power Socket near TRF. T-603	41465	1	41465	0	0	c
8	75kVA Dry Transformer **(refer to table # 35-4)**	75000	1	75000	0.5224	39180	h
9	48V Battery Charger for Communication	8180	1	8180	0.5	4090	a
10	Outdoor Lighting Panel (OLP)**(refer to table # 35-7)**	10425.6	1	10425.6	1	10425.6	
	Bus Section -2						
1	125V Battery Charger-2	37483	1	37483	0.5	18741.5	a
2	PACU-3 & 4 for 132kV GIS room	80000	2	160000	0.5	80000	a
3	Control panel for HVAC & Duct mounted Electric Heater	21176	1	21176	0.05	1058.8	g
4	EOT Crane at 132kV GIS Room	11580	1	11580	0	0	c
5	Power Panel -2 (PP-2) **(refer to table # 35- 3)**	3415.5	1	3415.5	1	3415.5	
6	380V Oil Filtration Receptacle -T-603	131640	1	131640	0	0	c
7	63 Amp Power Socket for PACUs	41465	2	82930	0.1	0	c
8	48V Battery Charger for Communication	8180	1	8180	0.5	4090	a
		Total				**306792**	

Table 35-2: Power Panel-1 Load

S.No.	Description of Load		Load per Equipment in VA (a)	Quantity (b)	Total Connected Load c = (a) X (b)	Diversity factor (d)	Operating Load (c) X (d)	Remarks Refer Clause 5
	380/220 V, 3Ph, 4wire Panel No:PP-1							
1	132kV GIS Room	220V Receptacle	180	5	900	0.5	450	b
2	13.8kV Switchgear Room	220V Receptacle	180	8	1440	0.5	720	b
3	Control Room	220V Receptacle	180	3	540	0.5	270	b
4	Record Room	220V Receptacle	180	2	360	0.5	180	b
5	Relay test room	220V Receptacle	180	2	360	0.5	180	b
6	Communication Room	220V Receptacle	180	3	540	0.5	270	b
7	Store Room	220V Receptacle	180	3	540	0.5	270	b
8	Toilet	220V Receptacle	180	2	360	0.5	180	b
9	Cable Basement	220V Receptacle	180	9	1620	0.5	810	
10	Portable Emergency Lighting	(2x35 Watts)	82	6	492	1	492	
11	Roof Mounted receptacles		180	1	180	1	180	
12	Power transformer 1 cooler fans		565	12	6780	1	6780	
13	Power transformer 2 cooler fans		565	12	6780	1	6780	
14	Power transformer 3 cooler fans		565	12	6780	1	6780	
15	Sump Pump at 132kV SWGR Basement		1000	2	2000	0	0	c
16	Booster Pump (Skid mounted)		1000	1	1000	0	0	c
17	Transfer Pump		1000	1	1000	0	0	c
18	OLTC Drive motor		4000	3	12000	0	0	c
19	Sump Pump in T601		1000	1	1000	0	0	c
20	Sump Pump in T602		1000	1	1000	0	0	c
21	Sump Pump in T603		1000	1	1000	0	0	c
22	Hand dryer		1500	1	1500	1	1500	

(CONTINUED) Table 35-2

	380/220 V, 3Ph, 4wire Panel No:PP-1							
23	Sump pump for SST		1000	2	2000	0	0	c
24	13.8kv Switchgear bus section -1	(Including future)	4350	1	4350	0.04	174	d&e
25	13.8kv Switchgear bus section -2		4350	1	4350	0.04	174	d&e
26	13.8kv Switchgear bus section -3	(Including future)	3750	1	3750	0.05	187.5	d&e
27	Power Transformer Loads consists of		441	3	1323	1	1323	
	Space Heater -(100VA)				0		0	
	Receptacles-180VA & Lighting - (15VA)				0		0	
	Tap changer control for relay (40VAx2)				0		0	
28	132kV GIS Local control Panels		357	9	3213	0.06	192.78	d&e
29	220V Receptacles for power transformer		180	3	540	0.5	270	b
30	220V Receptacles for power SST		180	2	360	0.5	180	b
		Total			**68058**		**28343.28**	

Table 35-3: Power Panel – 2 Load

S.No.	Description of Load	Load per Equipment in VA (a)	Quantity (b)	Total Connected Load c =(a) X (b)	Diversity factor (d)	Operating Load (c) X (d)	Remarks Refer Clause 5
	380/220 V, 3Ph, 4-wire Panel No:PP-2						
	220V Supply to following Control & Protection Panel						
1	132kV O/H Line Set-1 & Set-2 Panels	330	2	660	0.27	178.2	d&e
2	132kV O/H Line Set-1 & Set-2 Panels	330	2	660	0.27	178.2	d&e
3	132/13.8kV Transformer Set-1 protn.	330	3	990	0.18	178.2	d&e
4	132/13.8kV Transformer Set-2 protn. & set-1 Mech protn.	330	6	1980	0.09	178.2	d&e
5	132KV Busbar protection-Main Zone	330	4	1320	0.14	184.8	d&e
6	132KV Busbar protection-Check Zone	330	2	660	0.27	178.2	d&e
7	132KV Bus coupler / Bus section protection	330	2	660	0.27	178.2	d&e
8	132kV Breaker fail protection Panel	330	4	1320	0.14	184.8	d&e
9	Auto Recloser Panel	330	2	660	0.27	178.2	d&e
10	132kV Syn. Check Panel	330	3	990	0.18	178.2	d&e
11	132kV Breaker control Panel	330	7	2310	0.08	184.8	d&e
12	Annunciator Panel with M.Kiosk	330	3	990	0.18	178.2	d&e
13	Data retrieval Panel	330	1	330	0.55	181.5	d&e
14	13.8kV Under frequency Load shedding panel	330	3	990	0.18	178.2	d&e
15	13.8kV auto charge over scheme	330	1	330	0.55	181.5	d&e
16	132kVManual Syn. swing Panel	330	1	330	0.55	181.5	d&e
17	Transformer RTCC Panel	330	3	990	0.18	178.2	d&e
18	Line Set1 Panels Future	330	2	660	0.27	178.2	d&e
19	Line Set2 Panels Future	330	2	660	0.27	178.2	d&e
		Total		**17490**		**3415.5**	

Table 35-4 : Main Lighting Distribution Board Load

S.No.	Description of Load	Load per Equipment in VA (a)	Quantity (b)	Total Connected Load c =(a) X (b)	Diversity factor (d)	Operating Load = (c) X (d)	Remarks Refer Clause 5
	220/127V,3Ph,4wire PANEL No-MLDB						
	MLDB FEEDER NO-1						
1	Indoor Lighting Loads-LP1 **(refer to table # 35-5)**	22264	1	22264	1	22264	
	MLDB FEEDER NO-2						
1	Indoor Lighting Loads-LP2 **(refer to table # 35-6)**	16918.05	1	16918.05	1	16918.05	
		Total		**39182.05**		**39182.05**	

Conclusion:

The Nearest Standard rating of 75kVA is selected for Dry type Transformers. Hence, Dry type transformer of 75kVA Capacity is Adequate.

Table 35-5 : Lighing Panel -1 Load

S.No.	Description of Load		Load per Equipment in VA (a)	Quantity (b)	Total Connected Load c = (a)X(b)	Diversity factor (d)	Operating Load (c) X (d)	Remarks Refer to Clause 5
	220/127 V, 3Ph, 4wire LIGHTING PANEL No:LP1							
1	132kV GIS Room	(1x400W HPMV)	471	8	3768	1	3768	
		(2x36W FTL)	85	3	255	1	255	
2	13.8kV Switchgear Room	(2x36W FTL)	85	42	3570	1	3570	
3	Control Room	(3x36W FTL)	127	24	3048	1	3048	
4	Record Room	(3x36W FTL)	127	5	635	1	635	
5	Relay test room	(2x36W FTL)	85	6	510	1	510	
6	Communication Room	(3x36W FTL)	127	6	762	1	762	
7	Batter Room	(2x36W FTL)	85	4	340	1	340	
8	Store Room	(2x36W FTL)	85	6	510	1	510	
9	Toilet / Pipe Chase	(1x36W FTL)	42	3	126	1	126	
10	Stair case	(2x36W FTL)	85	5	425	1	425	
11	Cable Basement	(2x36W FTL)	85	50	4250	1	4250	
12	Building Perimeter Lghiting	(1x100W Lamp)	118	8	944	1	944	
		(1x60W Lamp)	71	6	426	1	426	
13	Emergency Lighting	(2x25W Lamp)	59	35	2065	1	2065	
14	Exit Sign Lighting Ckts	(2x18W Lamp)	42	15	630	1	630	
			Total		**22264**		**22264**	

Table 35-6 : Lighting Panel -2 Load

S.No.	Description of Load		Load per Equipment in VA (a)	Quantity (b)	Total Connected Load c =(a) X(b)	Diversity factor (d)	Operating Load (c) X (d)	Remarks Refer to Clause 5
	220/127 V, 3Ph, 4wire LIGHTING PANEL No:LP2							
1	132kV GIS Room	127V Receptacle	180	5	900	0.5	450	b
2	13.8kV Switchgear Room	127V Receptacle	180	8	1440	0.5	720	b
3	Control Room	127V Receptacle	180	3	540	0.5	270	b
4	Record Room	127V Receptacle	180	2	360	0.5	180	b
5	Relay test room	127V Receptacle	180	2	360	0.5	180	b
6	Communtion Room	127V Receptacle	180	3	540	0.5	270	b
7	Store Room	127V Receptacle	180	3	540	0.5	270	b
8	Toilet	127V Receptacle	180	1	180	1	180	
9	Cable Basement	127V Receptacle	180	9	1620	0.5	810	b
10	PACU Panel	127V Receptacle	180	1	180	1	180	
11	SCADA (RTU+IFC) Panels Supply		841	2	1682	1	1682	
12	SOE (RTU+IFC) Panels Supply		1364	2	2728	1	2728	
13	132kV O/H Line Set-1		247	2	494	0.5	247	d&f
14	132kV O/H Line Set-2		247	2	494	0.5	247	d&f
15	132/13.8kV Transformer Set-1 Protn.		247	3	741	0.33	244.53	d&f
16	132/13.8kV Transformer Set-2 Protn.		247	3	741	0.33	244.53	d&f
17	132/13.8kV Transformer Set-2 Mech Protn.		247	3	741	0.33	244.53	d&f
18	132/13.8kV Busbar Protection-Main Zone		247	4	988	0.25	247	d&f
19	132/13.8kV Busbar Protection-Check Zone		247	2	494	0.5	247	d&f
20	132kV Buscoupler / Bus section Protection		247	2	494	0.5	247	d&f

CONTINUE Table 35-6

21	132kV Breaker Fail Protection Panel	247	4	988	0.25	247	d&f
22	Auto Reclose Panel	247	2	494	0.5	247	d&f
23	132kV syn.Check Panel for OHL	247	1	247	1	247	d&f
24	132kV syn.Check Panel for Bus Coupler & Bus Section	247	1	247	1	247	d&f
25	132kV syn.Check Panel for Transformer	247	1	247	1	247	d&f
26	132kV Breaker Control Panel	247	7	1729	0.14	242.06	d&f
27	Annunciator Panel with M.Kiosk	247	3	741	0.33	244.53	d&f
28	Data retreival Panel	247	1	247	1	247	d&f
29	13.8kV Under frequancy Load shedding panel	247	3	741	0.33	244.53	d&f
30	13.8kV auto charge over scheme	247	1	247	1	247	d&f
31	132kV Manual Syn.swing Panel	247	1	247	1	247	d&f
32	Transformer RTCC Panel	247	3	741	0.33	244.53	d&f
33	Line Set1 Panels Future	247	2	494	0.5	247	d&f
34	Line Set2 Panels Future	247	2	494	0.5	247	d&f
35	GIS LCCs - Supply	419	9	3771	0.11	414.81	d&f
36	13.8kV SWGR Sec-1 Panel Ltg, Ventilation Fan, Socket	1326	1	1326	1	1326	
37	13.8kV SWGR Sec-2 Panel Ltg, Ventilation Fan, Socket	1304	1	1304	1	1304	
38	13.8kV SWGR Sec-3 Panel Ltg, Ventilation Fan, Socket	539	1	539	1	539	
		Total		**31101**		**16918.05**	

Table 35-7 : Outdoor Lighting Panel Load

S.No.	Description of Load	Load per Equipment in VA (a)	Quantity (b)	Total Connected Load c= (a)X(b)	Diversity factor (d)	Operating Load (c) X (d)	Remarks Refer to Clause 5e
	Outdoor LIGHTING Loads Details Panel :OLP						
1	Outdoor LIGHT fittings- 1x400W	471	18	8478	1	8478	d
2	Gate post Lighting -1x250W	294	4	1176	1	1176	d
3	Sign board lighting -1x250W	294	2	588	1	588	d
4	220V Receptacle near Lighting Mast (LM)	180	6	1080	0.17	183.6	d
		Total		**11322**		**10425.6**	

5.0 CALCULATIONS

Station Auxiliary Transformer Sizing for main AC distribution operating load is **306792**VA, therefore:

- The simultaneous peak load on Auxiliary Transformer excluding Oil Filtration Socket (OFS). **306792**VA
- The simultaneous peak load on Auxiliary Transformer including One Oil Filtration Socket (OFS). **(131640 + 306792) = 438432VA**
- Selected Auxiliary Transformer rating per 51-TMSS-01 (nearest size) **500 kVA**

The Size of Selected Station Transformer with capacity of 500 kVA is Adequate to meet the total calculated simultaneous AC loads as above.

6.0 BIBLIOGRAPHY

51-TMSS-01, “Pad Mounted Station Service Transformer”.